

[CHUNK_ID: 001]

[TOPIC: Jig Overview - Khai niem co ban]

[KEYWORDS: jig, preview, autocad .net api, interactive, drag, entity creation]

Jig La Gi? - Tong Quan Ve Jig Trong AutoCAD .NET API

Jig la co che trong AutoCAD .NET API cho phep hien thi preview dong cua entity khi nguoi dung di chuyen chuot. Entity thay doi hinh dang, vi tri, kich thuc theo thoi gian thuc truoc khi duoc them vao drawing database. Jig hoat dong bang cach: (1) Thu thap input tu chuot (diem, khoang cach, goc) trong phuong thuc Sampler(), (2) Cap nhat entity trong Update() hoac WorldDraw(), (3) Lap lai cho den khi nguoi dung click xac nhan hoac Cancel. AutoCAD .NET API cung cap 2 loai jig: EntityJig (jig 1 entity cu the) va DrawJig (jig bat ky do hoa nao, nhieu entity). Jig duoc goi thong qua Editor.Drag(jig) va tra ve PromptResult voi Status la OK, Cancel, hoac Keyword.

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 002]

[TOPIC: So sanh EntityJig va DrawJig]

[JIG_TYPE: EntityJig, DrawJig]

[KEYWORDS: EntityJig, DrawJig, so sanh, UCS, WCS, Sampler, Update, WorldDraw]

EntityJig vs DrawJig - Khi Nao Dung Loai Nao

EntityJig: ke thua tu Jig, truyen entity vao constructor. Override Sampler() va Update(). Chi jig duoc 1 entity. Diem trong Sampler() mac dinh la UCS. Don gian, phu hop cho jig 1 entity nhu Line, Circle, Arc, Solid3d. DrawJig: ke thua tu Jig, khong truyen entity. Override Sampler() va WorldDraw(). Tu ve do hoa trong WorldDraw() bang draw.Geometry.Draw(). Diem trong Sampler() la WCS (phai tu chuyen doi sang UCS). Jig duoc nhieu entity cung luc, hoac ve custom graphics. Phu hop cho move nhieu entity, jig voi dimension dong. Cay ke thua: Jig (abstract) -> EntityJig (Sampler + Update) | DrawJig (Sampler + WorldDraw).

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 003]

[TOPIC: Sampler Pattern - Thu thap input trong Jig]

[KEYWORDS: Sampler, SamplerStatus, JigPromptPointOptions, AcquirePoint, AcquireDistance, AcquireAngle, NoChange, flicker]

Sampler() - Pattern Thu Thap Input Trong Jig

Sampler() la phuong thuc duoc AutoCAD goi lien tuc trong vong lap jig de thu thap input tu nguoi dung. Cac phuong thuc acquire: prompts.AcquirePoint() (lay diem), prompts.AcquireDistance() (lay khoang cach), prompts.AcquireAngle() (lay goc), prompts.AcquireString() (lay chuo). SamplerStatus.OK: co input moi, AutoCAD se goi Update/WorldDraw. SamplerStatus.NoChange: input giong lan truoc, khong redraw (QUAN TRONG de tranh flicker). SamplerStatus.Cancel: nguoi dung nhan Escape. UserInputControls flags thuong dung: Accept3dCoordinates, GovernedByUCSDetect, NoZeroResponseAccepted, NoNegativeResponseAccepted, NullResponseAccepted, AcceptOtherInputString (cho keywords). Luon so sanh gia tri moi voi gia tri cu: if (res.Value.IsEqualTo(_prevPoint)) return SamplerStatus.NoChange; hoac if (Math.Abs(res.Value - _prevValue) < 1e-6) return SamplerStatus.NoChange;

```
protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptPointOptions opts = new
JigPromptPointOptions("\nChon diem: "); opts.UserInputControls = UserInputControls.Accept3dCoordinates |
UserInputControls.GovernedByUCSDetect; opts.BasePoint = _basePoint; opts.UseBasePoint = true;
PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return
SamplerStatus.Cancel; if (res.Value.IsEqualTo(_currentPoint)) return SamplerStatus.NoChange; _currentPoint =
res.Value; return SamplerStatus.OK; }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 004]

[TOPIC: Cach goi Jig tu Command method]

[KEYWORDS: CommandMethod, Editor.Drag, Transaction, BlockTableRecord, AppendEntity, PromptResult]

Goi Jig Tu Command - Pattern Co Ban

Quy trinh goi jig: (1) Lay Editor tu Document. (2) Mo Transaction. (3) Tao entity. (4) Tao jig instance, truyen entity vao. (5) Goi ed.Drag(jig). (6) Kiem tra PromptResult.Status. (7) Neu OK, them entity vao BlockTableRecord va Transaction. (8) Commit Transaction. QUAN TRONG: Chi them entity vao database SAU KHI jig thanh cong (PromptStatus.OK). Ngoai tru Block Reference voi Attributes - phai them vao DB truoc khi jig de attributes hien thi.

```
[CommandMethod("MYJIG")] public static void MyJigCommand() { Document doc = Application.DocumentManager.MdiActiveDocument; Database db = doc.Database; Editor ed = doc.Editor; PromptPointResult ptRes = ed.GetPoint("\nChon diem: "); if (ptRes.Status != PromptStatus.OK) return; using (Transaction tr = db.TransactionManager.StartTransaction()) { BlockTableRecord btr = (BlockTableRecord)tr.GetObject( db.CurrentSpaceId, OpenMode.ForWrite); Circle circle = new Circle(ptRes.Value, Vector3d.ZAxis, 1e-5); CircleJig jig = new CircleJig(circle, ptRes.Value); PromptResult res = ed.Drag(jig); if (res.Status == PromptStatus.OK) { btr.AppendEntity(circle); tr.AddNewlyCreatedDBObject(circle, true); } tr.Commit(); } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 005]
[TOPIC: Line Jig - EntityJig ve duong thang]
[ENTITY_TYPE: Line]
[JIG_TYPE: EntityJig]
[KEYWORDS: Line, EntityJig, StartPoint, EndPoint, AcquirePoint, rubber band]

Line Jig - Ve Duong Thang Voi EntityJig

Line Jig: nguoi dung chon diem dau (truoc khi jig), sau do keo chuot de xac dinh diem cuoi. Entity: Autodesk.AutoCAD.DatabaseServices.Line. Thuoc tinh can cap nhat: EndPoint. Diem dau (StartPoint) duoc set truoc khi jig, chi jig diem cuoi. Su dung AcquirePoint voi BasePoint = StartPoint, UseBasePoint = true de hien thi rubber band.

```
public class LineJig : EntityJig { private Point3d _startPoint; private Point3d _endPoint; public LineJig(Line line, Point3d startPoint) : base(line) { _startPoint = startPoint; _endPoint = startPoint; line.StartPoint = startPoint; line.EndPoint = startPoint; } protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptPointOptions opts = new JigPromptPointOptions("\nChon diem cuoi: "); opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.GovernedByUCSDetect; opts.BasePoint = _startPoint; opts.UseBasePoint = true; PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (res.Value.IsEqualTo(_endPoint)) return SamplerStatus.NoChange; _endPoint = res.Value; return SamplerStatus.OK; } protected override bool Update() { ((Line)Entity).EndPoint = _endPoint; return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 006]
[TOPIC: Circle Jig - EntityJig ve hinh tron]
[ENTITY_TYPE: Circle]
[JIG_TYPE: EntityJig]
[KEYWORDS: Circle, EntityJig, Radius, AcquireDistance, Center]

Circle Jig - Ve Hinh Tron Voi EntityJig

Circle Jig: nguoi dung chon tam (truoc jig), keo chuot de xac dinh ban kinh. Entity: Autodesk.AutoCAD.DatabaseServices.Circle. Thuoc tinh can cap nhat: Radius. Su dung AcquireDistance voi BasePoint = Center. Khoi tao Radius = 1e-5 (khong duoc = 0, se loi). So sanh: Math.Abs(res.Value - _radius) < Tolerance.Global.EqualPoint de tra ve NoChange.

```
public class CircleJig : EntityJig { private Point3d _center; private double _radius; public CircleJig(Circle circle, Point3d center) : base(circle) { _center = center; _radius = 0; circle.Center = center; circle.Radius = 1e-5; } protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptDistanceOptions opts = new JigPromptDistanceOptions("\nChon ban kinh: "); opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.NoZeroResponseAccepted | UserInputControls.NoNegativeResponseAccepted; opts.BasePoint = _center; opts.UseBasePoint = true; PromptDoubleResult res = prompts.AcquireDistance(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (Math.Abs(res.Value - _radius) < Tolerance.Global.EqualPoint) return SamplerStatus.NoChange; _radius = res.Value; return SamplerStatus.OK; } protected override bool Update() { if (_radius > 0) ((Circle)Entity).Radius = _radius; return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 007]
[TOPIC: Arc Jig - EntityJig ve cung tron (multi-phase)]
[ENTITY_TYPE: Arc]
[JIG_TYPE: EntityJig]
[KEYWORDS: Arc, EntityJig, multi-phase, AcquireDistance, AcquireAngle, Radius, StartAngle, EndAngle]

Arc Jig - Ve Cung Tron Voi Multi-Phase EntityJig

Arc Jig can 2 phase: Phase 0 chon ban kinh (AcquireDistance), Phase 1 chon goc ket thuc (AcquireAngle). Entity: Autodesk.AutoCAD.DatabaseServices.Arc. Thuoc tinh: Center, Radius, StartAngle, EndAngle. Tam duoc chon truoc jig. Dung bien _phase de phan biet tung buoc. Command goi: set jig.Phase = 0, Drag(); set jig.Phase = 1, Drag() lan 2.

```
public class ArcJig : EntityJig { private Point3d _center; private double _radius; private double _endAngle;
private int _phase; public int Phase { get { return _phase; } set { _phase = value; } } public ArcJig(Arc arc,
Point3d center) : base(arc) { _center = center; _radius = 1e-5; _endAngle = 0; _phase = 0; arc.Center =
center; arc.Radius = 1e-5; arc.StartAngle = 0; arc.EndAngle = Math.PI; } protected override SamplerStatus
Sampler(JigPrompts prompts) { if (_phase == 0) { JigPromptDistanceOptions opts = new
JigPromptDistanceOptions("\nChon ban kinh: "); opts.BasePoint = _center; opts.UseBasePoint = true;
opts.UserInputControls = UserInputControls.NoZeroResponseAccepted |
UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if
(res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (Math.Abs(res.Value - _radius) < 1e-6)
return SamplerStatus.NoChange; _radius = res.Value; return SamplerStatus.OK; } else { JigPromptAngleOptions
opts = new JigPromptAngleOptions("\nChon goc ket thuc: "); opts.BasePoint = _center; opts.UseBasePoint = true;
PromptDoubleResult res = prompts.AcquireAngle(opts); if (res.Status == PromptStatus.Cancel) return
SamplerStatus.Cancel; if (Math.Abs(res.Value - _endAngle) < 1e-6) return SamplerStatus.NoChange; _endAngle =
res.Value; return SamplerStatus.OK; } } protected override bool Update() { Arc arc = (Arc)Entity; if (_phase
== 0) { if (_radius > 0) arc.Radius = _radius; } else { arc.EndAngle = _endAngle; } return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 008]

[TOPIC: Ellipse Jig - EntityJig ve hinh ellipse (multi-phase)]

[ENTITY_TYPE: Ellipse]

[JIG_TYPE: EntityJig]

[KEYWORDS: Ellipse, EntityJig, MajorAxis, MinorRatio, Set, multi-phase]

Ellipse Jig - Ve Hinh Ellipse Voi Multi-Phase

Ellipse Jig can 2 phase: Phase 0 chon ban kinh lon (major radius), Phase 1 chon ban kinh nho (tinh ratio). Entity: Autodesk.AutoCAD.DatabaseServices.Ellipse. Su dung Ellipse.Set(center, normal, majorAxis, ratio, startAngle, endAngle). Ratio = minorRadius / majorRadius, phai <= 1.0. Ca 2 phase deu dung AcquireDistance voi BasePoint = center.

```
public class EllipseJig : EntityJig { private Point3d _center; private double _majorRadius; private double
_minorRatio; private int _phase; public int Phase { get { return _phase; } set { _phase = value; } } public
EllipseJig(Ellipse ellipse, Point3d center) : base(ellipse) { _center = center; _majorRadius = 1e-5;
_minorRatio = 0.5; _phase = 0; ellipse.Set(_center, Vector3d.ZAxis, new Vector3d(1e-5, 0, 0), 0.5, 0, 2 *
Math.PI); } protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptDistanceOptions opts = new
JigPromptDistanceOptions(_phase == 0 ? "\nChon ban kinh lon: " : "\nChon ban kinh nho: "); opts.BasePoint =
_center; opts.UseBasePoint = true; opts.UserInputControls = UserInputControls.NoZeroResponseAccepted |
UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if
(res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (_phase == 0) { if (Math.Abs(res.Value -
_majorRadius) < 1e-6) return SamplerStatus.NoChange; _majorRadius = res.Value; } else { double ratio =
Math.Min(res.Value / _majorRadius, 1.0); if (Math.Abs(ratio - _minorRatio) < 1e-6) return
SamplerStatus.NoChange; _minorRatio = ratio; } return SamplerStatus.OK; } protected override bool Update() {
((Ellipse)Entity).Set(_center, Vector3d.ZAxis, new Vector3d(_majorRadius, 0, 0), _minorRatio, 0, 2 * Math.PI);
return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 009]

[TOPIC: Polyline Jig - EntityJig ve polyline lien tuc (loop pattern)]

[ENTITY_TYPE: Polyline]

[JIG_TYPE: EntityJig]

[KEYWORDS: Polyline, EntityJig, AddVertexAt, SetPointAt, vertex loop, Convert2d, Plane, multi-vertex]

Polyline Jig - Ve Polyline Lien Tuc Voi Loop Pattern

Polyline Jig la jig phuc tap nhat trong 2D vi can them vertex lien tuc. Pattern: Tao Polyline voi 2 vertex (diem dau + diem tam). Moi lan Drag() thanh cong, goi AddVertex() de them vertex moi, roi Drag() tiep. Khi Cancel/Enter, dung loop. Loai bo vertex cuoi cung (vertex tam) truoc khi them vao DB. Su dung Point3d.Convert2d(Plane) de chuyen Point3d sang Point2d cho Polyline.AddVertexAt(). Plane mac dinh: new Plane(Point3d.Origin, Vector3d.ZAxis).

```
public class PolylineJig : EntityJig { private Point3d _tempPoint; private Plane _plane; public
PolylineJig(Polyline pline, Point3d start) : base(pline) { _plane = new Plane(Point3d.Origin, Vector3d.ZAxis);
_tempPoint = start; pline.AddVertexAt(0, start.Convert2d(_plane), 0, 0, 0); pline.AddVertexAt(1,
start.Convert2d(_plane), 0, 0, 0); } protected override SamplerStatus Sampler(JigPrompts prompts) {
JigPromptPointOptions opts = new JigPromptPointOptions("\nChon diem tiep theo: "); opts.UserInputControls =
UserInputControls.Accept3dCoordinates | UserInputControls.GovernedByUCSDetect |
UserInputControls.NullResponseAccepted; opts.BasePoint = _tempPoint; opts.UseBasePoint = true;
PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return
SamplerStatus.Cancel; if (res.Value.IsEqualTo(_tempPoint)) return SamplerStatus.NoChange; _tempPoint =
res.Value; return SamplerStatus.OK; } protected override bool Update() { Polyline pline = (Polyline)Entity;
pline.SetPointAt(pline.NumberOfVertices - 1, _tempPoint.Convert2d(_plane)); return true; } public void
AddVertex() { Polyline pline = (Polyline)Entity; pline.AddVertexAt(pline.NumberOfVertices,
_tempPoint.Convert2d(_plane), 0, 0, 0); } } // Command pattern: // while (true) { res = ed.Drag(jig); if OK ->
```

```
jig.AddVertex(); else break; } // Remove last temp vertex before adding to DB.
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 010]

[TOPIC: MText Jig - EntityJig dat vi tri van ban da dong]

[ENTITY_TYPE: MText]

[JIG_TYPE: EntityJig]

[KEYWORDS: MText, EntityJig, Location, TextHeight, Contents, AcquirePoint, text placement]

MText Jig - Dat Vi Tri Van Ban Da Dong

MText Jig: nhap noi dung text truoc (ed.GetString), sau do jig de chon vi tri dat text. Entity:

Autodesk.AutoCAD.DatabaseServices.MText. Thuoc tinh can cap nhat: Location (Point3d). Chi can 1 phase:

AcquirePoint cho vi tri. Co the mo rong thanh 2 phase: phase 0 chon vi tri, phase 1 chon chieu cao (TextHeight).

```
public class MTextJig : EntityJig { private Point3d _position; public MTextJig(MText mtext) : base(mtext) {
    _position = Point3d.Origin; mtext.Location = Point3d.Origin; } protected override SamplerStatus
    Sampler(JigPrompts prompts) { JigPromptPointOptions opts = new JigPromptPointOptions("\nChon vi tri dat text:
    "); opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.GovernedByUCSDetect;
    PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return
    SamplerStatus.Cancel; if (res.Value.IsEqualTo(_position)) return SamplerStatus.NoChange; _position =
    res.Value; return SamplerStatus.OK; } protected override bool Update() { ((MText)Entity).Location = _position;
    return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 011]

[TOPIC: DBText Jig - EntityJig dat vi tri va chieu cao text don dong (multi-phase)]

[ENTITY_TYPE: DBText]

[JIG_TYPE: EntityJig]

[KEYWORDS: DBText, EntityJig, Position, Height, TextString, multi-phase, AcquirePoint, AcquireDistance]

DBText Jig - Van Ban Don Dong Voi Multi-Phase

DBText Jig can 2 phase: Phase 0 chon vi tri (Position), Phase 1 chon chieu cao (Height). Entity:

Autodesk.AutoCAD.DatabaseServices.DBText. Noi dung (TextString) duoc nhap truoc khi jig. Phase 0: AcquirePoint.

Phase 1: AcquireDistance voi BasePoint = Position da chon.

```
public class DBTextJig : EntityJig { private Point3d _position; private double _height; private int _phase;
    public int Phase { get { return _phase; } set { _phase = value; } } public DBTextJig(DBText text, string
    content) : base(text) { _position = Point3d.Origin; _height = 2.5; _phase = 0; text.TextString = content;
    text.Height = _height; text.Position = Point3d.Origin; } protected override SamplerStatus Sampler(JigPrompts
    prompts) { if (_phase == 0) { JigPromptPointOptions opts = new JigPromptPointOptions("\nChon vi tri text: ");
    opts.UserInputControls = UserInputControls.Accept3dCoordinates; PromptPointResult res =
    prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if
    (res.Value.IsEqualTo(_position)) return SamplerStatus.NoChange; _position = res.Value; return
    SamplerStatus.OK; } else { JigPromptDistanceOptions opts = new JigPromptDistanceOptions("\nChon chieu cao: ");
    opts.BasePoint = _position; opts.UseBasePoint = true; opts.UserInputControls =
    UserInputControls.NoZeroResponseAccepted | UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult
    res = prompts.AcquireDistance(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if
    (Math.Abs(res.Value - _height) < 1e-6) return SamplerStatus.NoChange; _height = res.Value; return
    SamplerStatus.OK; } } protected override bool Update() { DBText text = (DBText)Entity; if (_phase == 0)
    text.Position = _position; else text.Height = _height; return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 012]

[TOPIC: Solid3d Box Jig - EntityJig tao hop 3D]

[ENTITY_TYPE: Solid3d]

[JIG_TYPE: EntityJig]

[KEYWORDS: Solid3d, Box, CreateBox, EntityJig, 3D, AcquireDistance, Displacement, TransformBy]

Solid3d Box Jig - Tao Hop 3D Voi EntityJig

Box Jig: chon tam, keo de xac dinh kich thuoc hop. Entity: Autodesk.AutoCAD.DatabaseServices.Solid3d. Su dung

Solid3d.CreateBox(length, width, height). CreateBox tao hop tai goc toa do, nen phai

TransformBy(Matrix3d.Displacement) de di chuyen ve tam da chon. Khoi tao voi CreateBox(1e-5, 1e-5, 1e-5) -

KHONG DUOC dung 0. Moi lan Update(), goi lai CreateBox() voi kich thuoc moi (CreateBox se thay the solid cu).

```
public class BoxJig : EntityJig { private Point3d _center; private double _size; public BoxJig(Solid3d solid,
    Point3d center) : base(solid) { _center = center; _size = 1e-5; } protected override SamplerStatus
```

```
Sampler(JigPrompts prompts) { JigPromptDistanceOptions opts = new JigPromptDistanceOptions("\nChon kich thuoc hop: "); opts.BasePoint = _center; opts.UseBasePoint = true; opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.NoZeroResponseAccepted | UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (Math.Abs(res.Value - _size) < 1e-6) return SamplerStatus.NoChange; _size = res.Value; return SamplerStatus.OK; } protected override bool Update() { Solid3d solid = (Solid3d)Entity; solid.CreateBox(_size, _size, _size); Vector3d disp = _center - Point3d.Origin; if (disp.Length > 1e-6) solid.TransformBy(Matrix3d.Displacement(disp)); return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 013]
[TOPIC: Solid3d Sphere Jig - EntityJig tao hình cầu 3D]
[ENTITY_TYPE: Solid3d]
[JIG_TYPE: EntityJig]
[KEYWORDS: Solid3d, Sphere, CreateSphere, EntityJig, 3D, Radius, Displacement]

Solid3d Sphere Jig - Tao Hình Cầu 3D

Sphere Jig: chọn tam, keo để xác định bán kính cầu. Sử dụng Solid3d.CreateSphere(radius). CreateSphere tạo tại gốc toạ độ, phải TransformBy để di chuyển. Chỉ cần 1 phase: AcquireDistance cho bán kính.

```
public class SphereJig : EntityJig { private Point3d _center; private double _radius; public SphereJig(Solid3d solid, Point3d center) : base(solid) { _center = center; _radius = 1e-5; } protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptDistanceOptions opts = new JigPromptDistanceOptions("\nChon ban kinh cau: "); opts.BasePoint = _center; opts.UseBasePoint = true; opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.NoZeroResponseAccepted | UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (Math.Abs(res.Value - _radius) < 1e-6) return SamplerStatus.NoChange; _radius = res.Value; return SamplerStatus.OK; } protected override bool Update() { Solid3d solid = (Solid3d)Entity; solid.CreateSphere(_radius); Vector3d disp = _center - Point3d.Origin; if (disp.Length > 1e-6) solid.TransformBy(Matrix3d.Displacement(disp)); return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 014]
[TOPIC: Solid3d Cylinder Jig - EntityJig tao hình trụ 3D (multi-phase)]
[ENTITY_TYPE: Solid3d]
[JIG_TYPE: EntityJig]
[KEYWORDS: Solid3d, Cylinder, CreateFrustum, EntityJig, 3D, multi-phase, Radius, Height]

Solid3d Cylinder Jig - Tao Hình Trụ 3D (2 Phase)

Cylinder Jig cần 2 phase: Phase 0 chọn bán kính, Phase 1 chọn chiều cao. Không có CreateCylinder() riêng, sử dụng CreateFrustum(height, xRadius, yRadius, topRadius) với xRadius = yRadius = topRadius. CreateFrustum tạo tại gốc toạ độ, phải TransformBy để di chuyển về center + offset Z.

```
public class CylinderJig : EntityJig { private Point3d _center; private double _radius; private double _height; private int _phase; public int Phase { get { return _phase; } set { _phase = value; } } public CylinderJig(Solid3d solid, Point3d center) : base(solid) { _center = center; _radius = 1e-5; _height = 1e-5; _phase = 0; } protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptDistanceOptions opts = new JigPromptDistanceOptions(_phase == 0 ? "\nChon ban kinh: " : "\nChon chieu cao: "); opts.BasePoint = _center; opts.UseBasePoint = true; opts.UserInputControls = UserInputControls.NoZeroResponseAccepted | UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; double oldVal = _phase == 0 ? _radius : _height; if (Math.Abs(res.Value - oldVal) < 1e-6) return SamplerStatus.NoChange; if (_phase == 0) _radius = res.Value; else _height = res.Value; return SamplerStatus.OK; } protected override bool Update() { Solid3d solid = (Solid3d)Entity; solid.CreateFrustum(_height, _radius, _radius, _radius); Vector3d disp = _center - Point3d.Origin; disp += new Vector3d(0, 0, _height / 2); if (disp.Length > 1e-6) solid.TransformBy(Matrix3d.Displacement(disp)); return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 015]
[TOPIC: Solid3d Frustum/Cone Jig - EntityJig tao non/non cut 3D (3 phase)]
[ENTITY_TYPE: Solid3d]
[JIG_TYPE: EntityJig]
[KEYWORDS: Solid3d, Frustum, Cone, CreateFrustum, EntityJig, 3D, multi-phase, bottomRadius, topRadius, height]

Solid3d Frustum/Cone Jig - Non Cut 3D (3 Phase)

Frustum là dạng tổng quát: bán kính đáy, bán kính đỉnh, chiều cao. Cone là trường hợp topRadius ~ 0. 3 phase: Phase 0 bán kính đáy, Phase 1 bán kính đỉnh, Phase 2 chiều cao. CreateFrustum(height, xRadius, yRadius, topRadius).

Pattern command: for (phase 0..2) { jig.Phase = phase; ed.Drag(jig); }

```
public class FrustumJig : EntityJig { private Point3d _center; private double _bottomRadius, _topRadius,
_height; private int _phase; public int Phase { get { return _phase; } set { _phase = value; } } public
FrustumJig(Solid3d solid, Point3d center) : base(solid) { _center = center; _bottomRadius = _topRadius =
_height = 1e-5; _phase = 0; } protected override SamplerStatus Sampler(JigPrompts prompts) { string[] msgs = {
"\nBan kinh day: ", "\nBan kinh dinh: ", "\nChieu cao: " }; JigPromptDistanceOptions opts = new
JigPromptDistanceOptions(msgs[_phase]); opts.BasePoint = _center; opts.UseBasePoint = true;
opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.NoZeroResponseAccepted |
UserInputControls.NoNegativeResponseAccepted; PromptDoubleResult res = prompts.AcquireDistance(opts); if
(res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; double oldVal = _phase == 0 ? _bottomRadius :
_phase == 1 ? _topRadius : _height; if (Math.Abs(res.Value - oldVal) < 1e-6) return SamplerStatus.NoChange;
switch (_phase) { case 0: _bottomRadius = res.Value; break; case 1: _topRadius = res.Value; break; default:
_height = res.Value; break; } return SamplerStatus.OK; } protected override bool Update() { Solid3d solid =
(Solid3d)Entity; solid.CreateFrustum(_height, _bottomRadius, _bottomRadius, _topRadius); Vector3d disp =
_center - Point3d.Origin; if (disp.Length > 1e-6) solid.TransformBy(Matrix3d.Displacement(disp)); return true;
} }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 016]

[TOPIC: DrawJig Move Entity - Di chuyen 1 entity voi preview]

[ENTITY_TYPE: Entity (bat ky)]

[JIG_TYPE: DrawJig]

[KEYWORDS: DrawJig, Move, WorldDraw, Geometry.Draw, Displacement, TransformBy, WCS, clone]

DrawJig Move Entity - Di Chuyen Voi Preview

DrawJig khác EntityJig: không truyền entity vào base constructor. Tu về do họa trong WorldDraw(). Diem trong Sampler() là WCS (không phải UCS). Pattern move: tính Vector3d displacement = currentPoint - basePoint, tạo clone entity, TransformBy(Matrix3d.Displacement), vẽ bằng draw.Geometry.Draw(clone), Dispose clone. SAU KHI jig OK, apply TransformBy lên entity gốc trong database.

```
public class MoveJig : DrawJig { private Entity _entity; private Point3d _basePoint; private Point3d
_prevPoint; private Point3d _currentPoint; public MoveJig(Entity entity, Point3d basePoint) { _entity =
entity; _basePoint = basePoint; _prevPoint = basePoint; _currentPoint = basePoint; } public Point3d
CurrentPoint => _currentPoint; protected override SamplerStatus Sampler(JigPrompts prompts) {
JigPromptPointOptions opts = new JigPromptPointOptions("\nChon diem dich: "); opts.BasePoint = _basePoint;
opts.UseBasePoint = true; opts.UserInputControls = UserInputControls.Accept3dCoordinates; PromptPointResult
res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if
(res.Value.IsEqualTo(_currentPoint)) return SamplerStatus.NoChange; _currentPoint = res.Value; return
SamplerStatus.OK; } protected override bool WorldDraw(WorldDraw draw) { Vector3d disp = _currentPoint -
_prevPoint; Matrix3d transform = Matrix3d.Displacement(disp); draw.Geometry.PushModelTransform(transform);
draw.Geometry.Draw(_entity); draw.Geometry.PopModelTransform(); _prevPoint = _currentPoint; return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 017]

[TOPIC: DrawJig Multi Entity Move - Di chuyen nhieu entity cung luc]

[ENTITY_TYPE: List]

[JIG_TYPE: DrawJig]

[KEYWORDS: DrawJig, multi entity, multiple, WorldDraw, Clone, TransformBy, GetSelection, move group]

DrawJig Multi Entity Move - Di Chuyen Nhieu Entity

DrawJig là cách duy nhất để jig nhiều entity cùng lúc (EntityJig chỉ hỗ trợ 1). Pattern: lưu List, trong WorldDraw() lặp qua từng entity, clone, TransformBy, Draw, Dispose. Command: GetSelection() -> lấy entities từ Transaction -> tạo jig -> Drag() -> apply TransformBy lên tất cả.

```
public class MultiEntityMoveJig : DrawJig { private List<Entity> _entities; private Point3d _basePoint;
private Point3d _currentPoint; public Vector3d Displacement => _currentPoint - _basePoint; public
MultiEntityMoveJig(List<Entity> entities, Point3d basePoint) { _entities = entities; _basePoint = basePoint;
_currentPoint = basePoint; } protected override SamplerStatus Sampler(JigPrompts prompts) {
JigPromptPointOptions opts = new JigPromptPointOptions("\nChon diem dich: "); opts.BasePoint = _basePoint;
opts.UseBasePoint = true; PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status ==
PromptStatus.Cancel) return SamplerStatus.Cancel; if (res.Value.IsEqualTo(_currentPoint)) return
SamplerStatus.NoChange; _currentPoint = res.Value; return SamplerStatus.OK; } protected override bool
WorldDraw(WorldDraw draw) { Vector3d disp = _currentPoint - _basePoint; Matrix3d transform =
Matrix3d.Displacement(disp); foreach (Entity ent in _entities) { Entity clone = (Entity)ent.Clone();
clone.TransformBy(transform); draw.Geometry.Draw(clone); clone.Dispose(); } return true; } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 018]

[TOPIC: DrawJig voi Dynamic Dimension - Hien thi kich thuoc dong khi jig]

[ENTITY_TYPE: Entity + AlignedDimension]

[JIG_TYPE: DrawJig]

[KEYWORDS: DrawJig, AlignedDimension, dynamic dimension, XLine1Point, XLine2Point, DimLinePoint, visual feedback]

DrawJig Voi Dynamic Dimension

DrawJig co the ve them AlignedDimension de hien thi khoang cach dong khi di chuyen. Tao AlignedDimension trong constructor. Trong WorldDraw(), set XLine1Point = basePoint, XLine2Point = currentPoint, DimLinePoint = trung diem + offset. draw.Geometry.Draw(dim) de ve dimension cung voi entity. Dimension khong duoc them vao DB, chi la transient graphics.

```
protected override bool WorldDraw(WorldDraw draw) { Vector3d disp = _currentPoint - _basePoint; Matrix3d transform = Matrix3d.Displacement(disp); Entity clone = (Entity)_entity.Clone(); clone.TransformBy(transform); draw.Geometry.Draw(clone); clone.Dispose(); _dim.XLine1Point = _basePoint; _dim.XLine2Point = _currentPoint; double offset = _basePoint.DistanceTo(_currentPoint) * 0.15; if (offset < 5) offset = 5; _dim.DimLinePoint = new Point3d( (_basePoint.X + _currentPoint.X) / 2, (_basePoint.Y + _currentPoint.Y) / 2 + offset, 0); draw.Geometry.Draw(_dim); return true; }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 019]

[TOPIC: Jig voi Keywords - Them tuy chon phim tat trong jig]

[KEYWORDS: Keywords, AcceptOtherInputString, PromptStatus.Keyword, JigPromptPointOptions, Arc, Close, Undo]

Jig Voi Keywords - Them Tuy Chon Trong Qua Trinh Jig

Keywords cho phép nguai dung nhap lenh phu trong khi jig (vi du: 'A' cho Arc, 'C' cho Close, 'U' cho Undo). Cach them: opts.Keywords.Add('Arc'); opts.AppendKeywordsToMessage = true; opts.UserInputControls |= UserInputControls.AcceptOtherInputString. Kiem tra: if (res.Status == PromptStatus.Keyword) { switch (res.StringResult) { ... } }. Khi nhan keyword, Sampler tra ve SamplerStatus.OK de trigger Update/WorldDraw xu ly logic moi.

```
protected override SamplerStatus Sampler(JigPrompts prompts) { JigPromptPointOptions opts = new JigPromptPointOptions("\nChon diem [Arc/Close/Undo]: "); opts.Keywords.Add("Arc"); opts.Keywords.Add("Close"); opts.Keywords.Add("Undo"); opts.AppendKeywordsToMessage = true; opts.UserInputControls = UserInputControls.Accept3dCoordinates | UserInputControls.AcceptOtherInputString; PromptPointResult res = prompts.AcquirePoint(opts); if (res.Status == PromptStatus.Keyword) { switch (res.StringResult) { case "Arc": _isArcMode = !_isArcMode; break; case "Close": _shouldClose = true; break; case "Undo": _shouldUndo = true; break; } return SamplerStatus.OK; } if (res.Status == PromptStatus.Cancel) return SamplerStatus.Cancel; if (res.Value.IsEqualTo(_currentPoint)) return SamplerStatus.NoChange; _currentPoint = res.Value; return SamplerStatus.OK; }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 020]

[TOPIC: Block Reference Jig voi Attributes - Phai them vao DB truoc khi jig]

[ENTITY_TYPE: BlockReference]

[JIG_TYPE: EntityJig]

[KEYWORDS: BlockReference, AttributeReference, AttributeDefinition, SetAttributeFromBlock, AppendEntity truoc jig, Erase neu cancel]

Block Reference Jig Voi Attributes

DIEM DAC BIET: BlockReference voi Attributes PHAI duoc them vao database TRUOC khi chay jig. Neu khong, attributes se khong hien thi trong preview. Quy trinh: (1) Tao BlockReference. (2) AppendEntity + AddNewlyCreatedDBObject. (3) Duyet BlockTableRecord de tim AttributeDefinition, tao AttributeReference tuong ung. (4) Chay jig. (5) Neu Cancel, goi bref.Erase() de xoa khoi DB. Day la ngoai le duy nhat cua quy tac 'chi them vao DB sau khi jig OK'.

```
[CommandMethod("BLOCKJIG")] public static void BlockJigCommand() { Document doc = Application.DocumentManager.MdiActiveDocument; Database db = doc.Database; Editor ed = doc.Editor; using (Transaction tr = db.TransactionManager.StartTransaction()) { BlockTable bt = (BlockTable)tr.GetObject(db.BlockTableId, OpenMode.ForRead); string blockName = "MyBlock"; if (!bt.Has(blockName)) return; BlockTableRecord blockDef = (BlockTableRecord)tr.GetObject( bt[blockName], OpenMode.ForRead); BlockTableRecord btr = (BlockTableRecord)tr.GetObject( db.CurrentSpaceId, OpenMode.ForWrite); BlockReference bref = new BlockReference(Point3d.Origin, bt[blockName]); btr.AppendEntity(bref); tr.AddNewlyCreatedDBObject(bref, true); foreach (ObjectId id in blockDef) { AttributeDefinition attDef = tr.GetObject(id, OpenMode.ForRead) as AttributeDefinition; if (attDef != null) { AttributeReference attRef = new AttributeReference(); attRef.SetAttributeFromBlock(attDef,
```

```
bref.BlockTransform()); bref.AttributeCollection.AppendAttribute(attRef); tr.AddNewlyCreatedDBObject(attRef, true); } } BlockJig jig = new BlockJig(bref); PromptResult res = ed.Drag(jig); if (res.Status != PromptStatus.OK) bref.Erase(); tr.Commit(); } }
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 021]

[TOPIC: Xu ly UCS va WCS trong Jig - Chuyen doi he toa do]

[KEYWORDS: UCS, WCS, CurrentUserCoordinateSystem, TransformBy, Inverse, Matrix3d, EntityJig UCS, DrawJig WCS]

Xu Ly UCS/WCS Trong Jig

EntityJig: Sampler() tra ve toa do UCS mac dinh (khi co GovernedByUCSDetect). DrawJig: Sampler() LUON tra ve toa do WCS. Phai tu chuyen doi sang UCS neu can. Chuyen doi: Matrix3d ucs = ed.CurrentUserCoordinateSystem; Matrix3d ucsInv = ucs.Inverse(); WCS -> UCS: point.TransformBy(ucsInv). UCS -> WCS: point.TransformBy(ucs). Khi lam viec voi Solid3d trong DrawJig, thuong can chuyen doi vi CreateBox/CreateSphere tao tai WCS origin.

```
Matrix3d ucs = ed.CurrentUserCoordinateSystem; Matrix3d ucsInverse = ucs.Inverse(); // WCS -> UCS Point3d ucsPoint = wcsPoint.TransformBy(ucsInverse); // UCS -> WCS Point3d wcsPoint = ucsPoint.TransformBy(ucs); // Trong DrawJig Sampler: PromptPointResult res = prompts.AcquirePoint(opts); // res.Value la WCS, chuyen ve UCS: Point3d ucsPoint = res.Value.TransformBy( ed.CurrentUserCoordinateSystem.Inverse());
```

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 022]

[TOPIC: Jig Best Practices va Tips - Cac luu y quan trong]

[KEYWORDS: best practices, tips, NoChange, flicker, 1e-5, Dispose, try-catch, cancel, visual style, UCS test]

Jig Best Practices Va Tips

1. LUON kiem tra NoChange: if (res.Value.IsEqualTo(_prev)) return SamplerStatus.NoChange - tranh flicker. 2. Khoi tao Solid3d voi 1e-5 thay vi 0 (CreateBox/CreateSphere khong chap nhan 0). 3. Chi AppendEntity SAU KHI jig OK, TRU BlockReference voi Attributes (phai them truoc, Erase neu cancel). 4. Boc Update()/WorldDraw() trong try-catch de tranh crash AutoCAD. 5. Test jig voi nhieu UCS khac nhau, khong chi WCS mac dinh. 6. Dispose entity neu jig cancel va entity chua trong DB. 7. EntityJig don gian hon cho 1 entity; DrawJig linh hoat hon cho nhieu entity hoac custom graphics. 8. Voi Solid3d, su dung visual style Conceptual/Realistic de thay ro 3D preview. 9. Polyline jig: loai bo vertex cuoi (vertex tam) truoc khi them vao DB. 10. Su dung BasePoint + UseBasePoint = true de hien thi rubber band line giua base va cursor.

--- CHUNK_BOUNDARY ---

[CHUNK_ID: 023]

[TOPIC: Bang tham khao nhanh - Entity va Jig tuong ung]

[KEYWORDS: reference table, entity type, jig type, acquire method, phases, Line, Circle, Arc, Ellipse, Polyline, MText, DBText, Solid3d, BlockReference, DrawJig]

Bang Tham Khao Nhanh - Jig Cho Tung Loai Entity

Line: EntityJig, AcquirePoint, 1 phase (endpoint). Circle: EntityJig, AcquireDistance, 1 phase (radius). Arc: EntityJig, AcquireDistance + AcquireAngle, 2-3 phases. Ellipse: EntityJig, AcquireDistance x2, 2 phases (major + ratio). Polyline: EntityJig, AcquirePoint loop, N phases (vertices). MText: EntityJig, AcquirePoint, 1 phase (position). DBText: EntityJig, AcquirePoint + AcquireDistance, 2 phases (position + height). Solid3d Box: EntityJig, AcquireDistance, 1-3 phases. Solid3d Sphere: EntityJig, AcquireDistance, 1 phase (radius). Solid3d Cylinder: EntityJig, AcquireDistance x2, 2 phases (radius + height). Solid3d Frustum/Cone: EntityJig, AcquireDistance x3, 3 phases. BlockReference: EntityJig, AcquirePoint, 1 phase (them vao DB truoc jig). Multi-Entity Move: DrawJig, AcquirePoint, 1 phase (clone + TransformBy trong WorldDraw). Move voi Dimension: DrawJig, AcquirePoint, 1 phase (ve AlignedDimension trong WorldDraw).